

Ch-2 Basics of HTML & Advance HTML5

HTML Introduction

- HTML stands for Hyper Text Markup Language
- HTML is the standard markup language for creating Web pages
- HTML describes the structure of a Web page
- HTML consists of a series of elements
- HTML elements tell the browser how to display the content
- HTML elements label pieces of content such as "this is a heading", "this is a paragraph", "this is a link", etc.

A Simple HTML Document

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

Explanation:

- The <!DOCTYPE html> declaration defines that this document is an HTML5 document
- The <html> element is the root element of an HTML page
- The <head> element contains meta information about the HTML page
- The <title> element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The <body> element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The <h1> element defines a large heading
- The <p> element defines a paragraph

What is an HTML Element?

An HTML element is defined by a start tag, some content, and an end tag:

<tagname> Content goes here... </tagname>

History of HTML

In the late 1980's, a physicist, Tim Berners-Lee who was a contractor at CERN, proposed a system for CERN researchers. In 1989, he wrote a memo proposing an internet based hypertext system.

Tim Berners-Lee is known as the father of HTML. The first available description of HTML was a document called "HTML Tags" proposed by Tim in late 1991. The latest version of HTML is HTML5.

Features of HTML

- 1) It is a very **easy and simple language**. It can be easily understood and modified.
- 2) It is very easy to make an **effective presentation** with HTML because it has a lot of formatting tags.
- 3) It is a **markup language**, so it provides a flexible way to design web pages along with the text.
- 4) It facilitates programmers to add a **link** on the web pages (by html anchor tag), so it enhances the interest of browsing of the user.
- 5) It is **platform-independent** because it can be displayed on any platform like Windows, Linux, and Macintosh, etc.
- 6) It facilitates the programmer to add **Graphics, Videos, and Sound** to the web pages which makes it more attractive and interactive.
- 7) HTML is a case-insensitive language, which means we can use tags either in lower-case or upper-case.

HTML Tags

HTML tags are like keywords which defines that how web browser will format and display the content. With the help of tags, a web browser can distinguish between an HTML content and a simple content. HTML tags contain three main parts: opening tag, content and closing tag. But some HTML tags are unclosed tags.

When a web browser reads an HTML document, browser reads it from top to bottom and left to right. HTML tags are used to create HTML documents and render their properties. Each HTML tags have different properties.

An HTML file must have some essential tags so that web browser can differentiate between a simple text and HTML text. You can use as many tags you want as per your code requirement.

- All HTML tags must enclosed within < > these brackets.
- Every tag in HTML perform different tasks.
- If you have used an open tag <tag>, then you must use a close tag </tag> (except some tags)

Syntax

<tag> content </tag>

HTML Attribute

- HTML attributes are special words which provide additional information about the elements or attributes are the modifier of the HTML element.
- Each element or tag can have attributes, which defines the behaviour of that element.
- Attributes should always be applied with start tag.
- The Attribute should always be applied with its name and value pair.
- The Attributes name and values are case sensitive, and it is recommended by W3C that it should be written in Lowercase only.
- You can add multiple attributes in one HTML element, but need to give space between two attributes.

Syntax

<element attribute_name="value">content</element>

HTML - Document Tags

HTML

The first and last tags in a document should always be the HTML tags. These are the tags that tell a Web browser where the HTML in your document begins and ends. Everything in the document goes inside <HTML> and </HTML>. This is a non-empty tag.

HEAD

The HEAD tags contain all of the document's header information. When I say "header," I don't mean what appears at the top of the browser window, but like the document title. HEAD section contains the information about the document. This information is communicated through the <TITLE> tag. This is a non-empty tag.

TITLE

This container is placed within the HEAD structure. Between the TITLE tags, you should have the title of your document. This will appear at the top of the browser's title bar. What you type should probably be something which indicates the document's contents. This is a non-empty tag.

BODY

This tag identifies the main portion of the HTML document. BODY comes after the HEAD structure. Between the BODY tags, you find all of the stuff that gets displayed in the browser window. All of the text, the graphics, and links, and so on these things occur between the BODY tags. This is a non-empty tag.

STRUCTURE OF HTML DOCUMENT :

A typical HTML document will have the following structure:

Document declaration tag

<html>

<head>

Document header related tags

<title>

Title of document

</title>

</head>

<body>

Document body related tags

</body>

</html>

EXAMPLE :

<HTML>

<HEAD>

```
<TITLE>DEMO</TITLE>  
</HEAD>
```

```
<BODY>  
    HELLO WORLD  
</BODY>  
</HTML>
```

HTML **<body>** Tag

The **<body>** tag defines the document's body.

The **<body>** element contains all the contents of an HTML document, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.

Syntax

```
<body> Place your Content here.....</body>
```

Attribute

Various attributes of BODY tag are as follows:

alink: This attribute specifies the color of hyperlinks as they being clicked. Set it to a color value or a predefined color name.

link: This attribute specifies the color of hyperlinks that have not yet been visited. Set it to a color value or a predefined color name.

vlink: This attribute specifies the color of hyperlinks that have been visited. Set it to a color value or a predefined color name.

background: This attribute specifies the url of the image or graphic file to be used in tiling the browser's background.

bgcolor: This attribute specifies the color of the browser as a background.

bgproperties: This attribute indicates if the background should scroll when the text does. If set to FIXED, the background will not scroll, only text will scroll.

bottommargin: This attribute specifies the bottom margin, the empty space at the bottom of document in pixels.

topmargin: This attribute specifies the top margin, the empty space at the top of document in pixels.

leftmargin: This attribute specifies the left margin, the empty space at the left of document in pixels.

rightmargin: This attribute specifies the right margin, the empty space at the right of document in pixels.

scroll: This attribute specifies whether a vertical scrollbar appears to the right of the document. Possible values are Yes (default) and No.

Example:

```
<HTML>
<HEAD>
<TITLE>Body Tag Attributes</TITLE>
</HEAD>
<BODY alink="red" vlink="pink" link="green"
background="E:\Neeta\Images\vacuoles.jpg" bgcolor="blue"
bgproperties="fixed" bottommargin=100 topmargin=200 leftmargin=250
rightmargin=300 scroll=no>
<H1>Hello World</H1>
</BODY>
</HTML>
```

Comment Tags

If you want to leave yourself notes in an HTML document, but don't want those notes to show up in the browser, you need to use the comment tag. To do that, use following:

```
<!-- textcomment -->
```

Example:

```
<!-- Hi, I'm a comment. -->
```

HTML Basic

HTML Headings

A HTML heading or HTML h tag can be defined as a title or a subtitle which you want to display on the webpage.

HTML headings are defined with the <h1> to <h6> tags.

<h1> defines the most important heading. <h6> defines the least important heading:

Alignment of Headings is done using **align** attribute. Possible values are right, center, justify and left.

Example

```
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
<h4>This is heading 4</h4>
<h5>This is heading 5</h5>
<h6>This is heading 6</h5>
```

HTML Paragraphs

HTML paragraph or HTML p tag is used to define a paragraph in a webpage. A paragraph always starts on a new line, and is usually a block of text.

HTML paragraphs are defined with the <p> tag. A paragraph always starts on a new line, and browsers automatically add some white space (a margin) before and after a paragraph.

Example

```
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>
```

Line Break Tag

Whenever you use the
 element, anything following it starts from the next line. This tag is an example of an empty element, where you do not need opening and closing tags, as there is nothing to go in between them. The
 tag has a space between the characters br and the forward slash. If you omit this space, older browsers will have trouble rendering the line break, while if you miss the forward slash character and just use
 it is not valid in XHTML.

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Line Break Example</title>
</head>
<body>
<p>Hello<br />
```

```

You delivered your assignment on time.<br />
Thanks<br />
Mahnaz</p>
</body>
</html>

```

Centering Content

You can use **<center>** tag to put any content in the center of the page or any table cell.

Example

```

<!DOCTYPE html>
<html>
<head>
<title>Centring Content Example</title>
</head>
<body>
<p>This text is not in the center.</p>
<center>
<p>This text is in the center.</p>
</center>
</body>
</html>

```

Horizontal Lines

Horizontal lines are used to visually break-up sections of a document. The **<hr>** tag creates a line from the current position in the document to the right margin and breaks the line accordingly. HTML

For example, you may want to give a line between two paragraphs as in the given example below:

Example

```

<!DOCTYPE html>
<html>
<head>
<title>Horizontal Line Example</title>
</head>
<body>
<p>This is paragraph one and should be on top</p>
<hr />
<p>This is paragraph two and should be at bottom</p>
</body>
</html>

```

Preserve Formatting

Sometimes, you want your text to follow the exact format of how it is written in the HTML document. In these cases, you can use the preformatted tag **<pre>**.

Any text between the opening **<pre>** tag and the closing **</pre>** tag will preserve the formatting of the source document.

Example

```
<!DOCTYPE html>
<html>
<head>
<title>Preserve Formatting Example</title>
</head>
<body>
<pre>
function testFunction( strText ){
alert (strText)
}
</pre>
</body>
</html>
```

HTML - Formatting

Bold Text

Anything that appears within **...** element, is displayed in bold

Example

```
<b>bold</b>
```

Italic Text

Anything that appears within **<i>...</i>** element is displayed in italicized

Example

```
<i>italicized</i>
```

Underlined Text

Anything that appears within **<u>...</u>** element, is displayed with underline

Example

```
<u>underlined</u>
```

Strike Text

Anything that appears within **<strike>...</strike>** element is displayed with strikethrough, which is a thin line through the text

Example

~~strikethrough~~

Superscript Text

The content of a **^{...}** element is written in superscript; the font size used is the same size as the characters surrounding it but is displayed half a character's height above the other characters.

Example

The following word uses a **^{superscript}** typeface.

Subscript Text

The content of a **_{...}** element is written in subscript; the font size used is the same as the characters surrounding it, but is displayed half a character's height beneath the other characters.

Example

The following word uses a **_{subscript}** typeface.

Inserted Text

Anything that appears within **<ins>...</ins>** element is displayed as inserted text.

Example

I want to drink ~~cola~~ wine

Deleted Text

Anything that appears within **...** element, is displayed as deleted text.

Example

I want to drink ~~cola~~ wine

Larger Text

The content of the **<big>...</big>** element is displayed one font size larger than the rest of the text surrounding it

Example

The following word uses a **<big>big</big>** typeface.

Smaller Text

The content of the **<small>...</small>** element is displayed one font size smaller than the rest of the text surrounding it.

Example

<p>The following word uses a <small>small</small> typeface.</p>

Grouping Content

The <div> and elements allow you to group together several elements to create sections or subsections of a page.

For example, you might want to put all of the footnotes on a page within a <div> element to indicate that all of the elements within that <div> element relate to the footnotes. You might then attach a style to this <div> element so that they appear using a special set of style rules.

Example

```

<body>
  <div id = "menu" align = "middle" >
    <a href = "/index.htm">HOME</a> |
    <a href = "/about/contact_us.htm">CONTACT</a> |
    <a href = "/about/index.htm">ABOUT</a>
  </div>

  <div id = "content" align = "left" >
    <h5>Content Articles</h5>
    <p>Actual content goes here.....</p>
  </div>

```

HTML Anchor

The **HTML anchor tag** defines *a hyperlink that links one page to another page*. It can create hyperlink to other web page as well as files, location, or any URL. The "href" attribute is the most important attribute of the HTML a tag. and which links to destination page or URL.

href attribute of HTML anchor tag

The href attribute is used to define the address of the file to be linked. In other words, it points out the destination page.

The syntax of HTML anchor tag is given below.

 Link Text

Example:

 this-link

Specify a location for Link using target attribute

By default, the linked page will be displayed in the current browser window. To change this, you must specify another target for the link.

The target attribute specifies where to open the linked document.

The target attribute can have one of the following values:

- `_self` - Default. Opens the document in the same window/tab as it was clicked
- `_blank` - Opens the document in a new window or tab
- `_parent` - Opens the document in the parent frame
- `_top` - Opens the document in the full body of the window

Example:

```
<a href="https://www.javatpoint.com/" target="_blank"> this-link </a>
```

Appearance of HTML anchor tag

An **unvisited link** is displayed underlined and blue.

A **visited link** displayed underlined and purple.

An **active link** is underlined and red.

HTML Images

Images can improve the design and the appearance of a web page.

The HTML `<img>` tag is used to embed an image in a web page.

Images are not technically inserted into a web page; images are linked to web pages. The `<img>` tag creates a holding space for the referenced image.

The `<img>` tag is empty, it contains attributes only, and does not have a closing tag.

Attributes of HTML img tag

The `src` and `alt` are important attributes of HTML `img` tag. All attributes of HTML image tag are given below.

1) `src`

It is a necessary attribute that describes the source or path of the image. It instructs the browser where to look for the image on the server.

The location of image may be on the same directory or another server.

2) alt

The alt attribute defines an alternate text for the image, if it can't be displayed. The value of the alt attribute describe the image in words. The alt attribute is considered good for SEO prospective.

3) width

It is an optional attribute which is used to specify the width to display the image. It is not recommended now. You should apply CSS in place of width attribute.

4) height

It h3 the height of the image. The HTML height attribute also supports iframe, image and object elements. It is not recommended now. You should apply CSS in place of height attribute.

Syntax

```

```

Example

```

```

HTML Tables

HTML tables allow web developers to arrange data into rows and columns.

We can create a table to display data in tabular form, using <table> element.

Table Rows

Each table row starts with a <tr> and ends with a </tr> tag.

tr stands for table row.

Table Cells

Each table cell is defined by a <td> and a </td> tag.

td stands for table data.

Everything between <td> and </td> are the content of the table cell.

Table Headers

Sometimes you want your cells to be table header cells. In those cases use the <th> tag instead of the <td> tag:

th stands for table header.

Table Caption

<caption> is used to define table caption.

Example:

```
<table>
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Smith</td>
    <td>43</td>
  </tr>
  <tr>
    <td>Jackson</td>
    <td>57</td>
  </tr>
</table>
```

Table Attributes

Cellpadding and Cellspacing Attributes

There are two attributes called *cellpadding* and *cellspacing* which you will use to adjust the white space in your table cells. The cellspacing attribute defines space between table cells, while cellpadding represents the distance between cell borders and the content within a cell.

Tables Backgrounds

You can set table background using one of the following two ways –

- **bghcolor** attribute – You can set background color for whole table or just for one cell.
- **background** attribute – You can set background image for whole table or just for one cell.

Table Height and Width

You can set a table width and height using **width** and **height** attributes. You can specify table width or height in terms of pixels or in terms of percentage of available screen area.

Colspan and Rowspan Attributes

You will use **colspan** attribute if you want to merge two or more columns into a single column. Similar way you will use **rowspan** if you want to merge two or more rows.

Example : Colspan

```
<table>
<tr>
  <th colspan="2">Name</th>
  <th>Age</th>
</tr>
<tr>
  <td>Jill</td>
  <td>Smith</td>
  <td>43</td>
</tr>
<tr>
  <td>Eve</td>
  <td>Jackson</td>
  <td>57</td>
</tr>
</table>
```

Example : Rowspan

```
<table>
<tr>
  <th>Name</th>
  <td>Jill</td>
</tr>
<tr>
  <th rowspan="2">Phone</th>
  <td>555-1234</td>
</tr>
<tr>
  <td>555-8745</td>
</tr>
</table>
```

HTML Lists

HTML lists allow web developers to group a set of related items in lists. All lists may contain one or more list elements.

There are three different types of HTML lists:

1. Ordered List or Numbered List (ol)
2. Unordered List or Bulleted List (ul)
3. Description List or Definition List (dl)

HTML Unordered Lists

In HTML Unordered list, all the list items are marked with bullets. It is also known as bulleted list also. The Unordered list starts with `<ul>` tag and list items start with the `<li>` tag.

The type Attribute

You can use **type** attribute for `<ul>` tag to specify the type of bullet you like. By default, it is a disc. Following are the possible options –

```
<ul type = "square">
<ul type = "disc">
<ul type = "circle">
```

Example

```
<ul>
  <li>Beetroot</li>
  <li>Ginger</li>
  <li>Potato</li>
  <li>Radish</li>
</ul>
```

HTML Ordered Lists

The HTML `<ol>` tag defines an ordered list. An ordered list can be numerical or alphabetical.

An ordered list starts with the `<ol>` tag. Each list item starts with the `<li>` tag.

The list items will be marked with numbers by default.

type Attribute

You can use **type** attribute for `<ol>` tag to specify the type of numbering you like. By default, it is a number. Following are the possible options –

```
<ol type = "1"> - Default-Case Numerals.
<ol type = "I"> - Upper-Case Numerals.
<ol type = "i"> - Lower-Case Numerals.
<ol type = "A"> - Upper-Case Letters.
<ol type = "a"> - Lower-Case Letters.
```

start Attribute

You can use **start** attribute for tag to specify the starting point of numbering you need. Following are the possible options –

<ol type = "1" start = "4"> - Numerals starts with 4.
 <ol type = "I" start = "4"> - Numerals starts with IV.
 <ol type = "i" start = "4"> - Numerals starts with iv.
 <ol type = "a" start = "4"> - Letters starts with d.
 <ol type = "A" start = "4"> - Letters starts with D.

Example

```
<ol>
  <li>Aries</li>
  <li>Bingo</li>
  <li>Leo</li>
  <li>Oracle</li>
</ol>
```

HTML Definition Lists

HTML and XHTML supports a list style which is called **definition lists** where entries are listed like in a dictionary or encyclopedia. The definition list is the ideal way to present a glossary, list of terms, or other name/value list.

Definition List makes use of following three tags.

- <dl> – Defines the start of the list
- <dt> – A term
- <dd> – Term definition
- </dl> – Defines the end of the list

Example

```
<dl>
  <dt><b>HTML</b></dt>
  <dd>This stands for Hyper Text Markup Language</dd>
  <dt><b>HTTP</b></dt>
  <dd>This stands for Hyper Text Transfer Protocol</dd>
</dl>
```

HTML – Frames

HTML frames are used to divide your browser window into multiple sections where each section can load a separate HTML document. A

collection of frames in the browser window is known as a frameset. The window is divided into frames in a similar way the tables are organized: into rows and columns.

Disadvantages of Frames

There are few drawbacks with using frames, so it's never recommended to use frames in your webpages –

- Some smaller devices cannot cope with frames often because their screen is not big enough to be divided up.
- Sometimes your page will be displayed differently on different computers due to different screen resolution.
- The browser's *back* button might not work as the user hopes.
- There are still few browsers that do not support frame technology.

Creating Frames

To use frames on a page we use `<frameset>` tag instead of `<body>` tag. The `<frameset>` tag defines, how to divide the window into frames. The **rows** attribute of `<frameset>` tag defines horizontal frames and **cols** attribute defines vertical frames. Each frame is indicated by `<frame>` tag and it defines which HTML document shall open into the frame.

Note – The `<frame>` tag deprecated in HTML5. Do not use this element.

Example

```
<html>

<head>
  <title>HTML Frames</title>
</head>

<frameset rows = "10%,80%,10%">
  <frame name = "top" src = "/html/top_frame.htm" />
  <frame name = "main" src = "/html/main_frame.htm" />
  <frame name = "bottom" src = "/html/bottom_frame.htm" />

  <noframes>
    <body>Your browser does not support frames.</body>
  </noframes>

</frameset>
```

</html>

The <frameset> Tag Attributes

Following are important attributes of the <frameset> tag –

Sr.No	Attribute & Description
	cols Specifies how many columns are contained in the frameset and the size of each column. You can specify the width of each column in one of the four ways – Absolute values in pixels. For example, to create three vertical frames, use <i>cols = "100, 500, 100"</i>. A percentage of the browser window. For example, to create three vertical frames, use <i>cols = "10%, 80%, 10%"</i>. Using a wildcard symbol. For example, to create three vertical frames, use <i>cols = "10%, *, 10%"</i>. In this case wildcard takes remainder of the window. As relative widths of the browser window. For example, to create three vertical frames, use <i>cols = "3*, 2*, 1*"</i>. This is an alternative to percentages. You can use relative widths of the browser window. Here the window is divided into sixths: the first column takes up half of the window, the second takes one third, and the third takes one sixth. rows This attribute works just like the cols attribute and takes the same values, but it is used to specify the rows in the frameset. For example, to create two horizontal frames, use <i>rows = "10%, 90%"</i>. You can specify the height of each row in the same way as explained above for columns. border This attribute specifies the width of the border of each frame in pixels. For example, <i>border = "5"</i>. A value of zero means no border. frameborder

This attribute specifies whether a three-dimensional border should be displayed between frames. This attribute takes value either 1 (yes) or 0 (no). For example `frameborder = "0"` specifies no border.

framespacing

- 5 This attribute specifies the amount of space between frames in a frameset. This can take any integer value. For example `framespacing = "10"` means there should be 10 pixels spacing between each frames.

The <frame> Tag Attributes

Following are the important attributes of <frame> tag –

Sr.No	Attribute & Description
	src
1	This attribute is used to give the file name that should be loaded in the frame. Its value can be any URL. For example, <code>src = "/html/top_frame.htm"</code> will load an HTML file available in html directory.
	name
2	This attribute allows you to give a name to a frame. It is used to indicate which frame a document should be loaded into. This is especially important when you want to create links in one frame that load pages into an another frame, in which case the second frame needs a name to identify itself as the target of the link.
	frameborder
3	This attribute specifies whether or not the borders of that frame are shown; it overrides the value given in the <code>frameborder</code> attribute on the <frameset> tag if one is given, and this can take values either 1 (yes) or 0 (no).
	marginwidth
4	This attribute allows you to specify the width of the space between the left and right of the frame's borders and the frame's content. The value is given in pixels. For example <code>marginwidth = "10"</code> .
5	marginheight

This attribute allows you to specify the height of the space between the top and bottom of the frame's borders and its contents. The value is given in pixels. For example `marginheight = "10"`.

noresize

- 6 By default, you can resize any frame by clicking and dragging on the borders of a frame. The `noresize` attribute prevents a user from being able to resize the frame. For example `noresize = "noresize"`.

scrolling

- 7 This attribute controls the appearance of the scrollbars that appear on the frame. This takes values either "yes", "no" or "auto". For example `scrolling = "no"` means it should not have scroll bars.

longdesc

- 8 This attribute allows you to provide a link to another page containing a long description of the contents of the frame. For example `longdesc = "framedescription.htm"`

HTML - Fonts

Fonts play a very important role in making a website more user friendly and increasing content readability. Font face and color depends entirely on the computer and browser that is being used to view your page but you can use HTML **** tag to add style, size, and color to the text on your website. You can use a **<basefont>** tag to set all of your text to the same size, face, and color.

The font tag is having three attributes called **size**, **color**, and **face** to customize your fonts. To change any of the font attributes at any time within your webpage, simply use the `<font>` tag. The text that follows will remain changed until you close with the `</font>` tag. You can change one or all of the font attributes within one `<font>` tag.

Set Font Size

You can set content font size using **size** attribute. The range of accepted values is from 1(smallest) to 7(largest). The default size of a font is 3.

```
<font size = "2">Font size = "2"</font><br />
```

Relative Font Size

You can specify how many sizes larger or how many sizes smaller than the preset font size should be. You can specify it like **** or ****

```
<font size = "-1">Font size = "-1"</font><br />
```

Setting Font Face

You can set font face using *face* attribute but be aware that if the user viewing the page doesn't have the font installed, they will not be able to see it. Instead user will see the default font face applicable to the user's computer.

```
<font face = "Times New Roman" size = "5">Times New Roman</font><br />
```

Setting Font Color

You can set any font color you like using *color* attribute. You can specify the color that you want by either the color name or hexadecimal code for that color.

```
<font color = "#FF00FF">This text is in pink</font><br />
```

The <basefont> Element

The <basefont> element is supposed to set a default font size, color, and typeface for any parts of the document that are not otherwise contained within a tag. You can use the elements to override the <basefont> settings.

The <basefont> tag also takes color, size and face attributes and it will support relative font setting by giving size a value of +1 for a size larger or -2 for two sizes smaller.

```
<basefont face = "arial, verdana, sans-serif" size = "2" color = "#ff0000">
  <p>This is the page's default font.</p>
  <h2>Example of the &lt;basefont&gt; Element</h2>
```

```
<p><font size = "+2" color = "darkgray">
  This is darkgray text with two sizes larger
</font>
</p>
```

```
<p><font face = "courier" size = "-1" color = "#000000">
```

It is a courier font, a size smaller and black in color.

</p>

HTML - Marquees

An HTML marquee is a scrolling piece of text displayed either horizontally across or vertically down your webpage depending on the settings. This is created by using HTML <marquees> tag.

Syntax

<marquee attribute_name = "attribute_value"....more attributes>

One or more lines or text message or image

</marquee>

The <marquee> Tag Attributes

Following is the list of important attributes which can be used with <marquee> tag.

Sr.No	Attribute & Description
	width
1	This specifies the width of the marquee. This can be a value like 10 or 20% etc.
	height
2	This specifies the height of the marquee. This can be a value like 10 or 20% etc.
	direction
3	This specifies the direction in which marquee should scroll. This can be a value like <i>up</i> , <i>down</i> , <i>left</i> or <i>right</i> .
	behavior
4	This specifies the type of scrolling of the marquee. This can have a value like <i>scroll</i> , <i>slide</i> and <i>alternate</i> .
	scrolldelay
5	This specifies how long to delay between each jump. This will have a value like 10 etc.

scrollamount

- 6 This specifies the speed of marquee text. This can have a value like 10 etc.

loop

- 7 This specifies how many times to loop. The default value is INFINITE, which means that the marquee loops endlessly.

bgcolor

- 8 This specifies background color in terms of color name or color hex value.

hspace

- 9 This specifies horizontal space around the marquee. This can be a value like 10 or 20% etc.

vspace

- 10 This specifies vertical space around the marquee. This can be a value like 10 or 20% etc.

Examples

```
<marquee>This is basic example of marquee</marquee>
```

HTML - Forms

An **HTML form** is a section of a document which contains controls such as text fields, password fields, checkboxes, radio buttons, submit button, menus etc.

An HTML form facilitates the user to enter data that is to be sent to the server for processing such as name, email address, password, phone number, etc. .

Syntax

```
<form action="server url" method="get|post">
  //input controls e.g. textfield, textarea, radiobutton, button
</form>
```

HTML Form Controls

There are different types of form controls that you can use to collect data using HTML form

- Text Input Controls
- Checkboxes Controls
- Radio Box Controls
- Select Box Controls
- File Select boxes
- Hidden Controls
- Clickable Buttons
- Submit and Reset Button

Text Input Controls

There are three types of text input used on forms –

- **Single-line text input controls** – This control is used for items that require only one line of user input, such as search boxes or names. They are created using HTML **<input>** tag.
- **Password input controls** – This is also a single-line text input but it masks the character as soon as a user enters it. They are also created using HTML **<input>** tag.
- **Multi-line text input controls** – This is used when the user is required to give details that may be longer than a single sentence. Multi-line input controls are created using HTML **<textarea>** tag.

Single-line text input controls

This control is used for items that require only one line of user input, such as search boxes or names. They are created using HTML **<input>** tag.

Attributes

Sr.No	Attribute & Description
	type
1	Indicates the type of input control and for text input control it will be set to text .
	name
2	Used to give a name to the control which is sent to the server to be recognized and get the value.

- 3 **value**
This can be used to provide an initial value inside the control.
- 4 **size**
Allows to specify the width of the text-input control in terms of characters.
- 5 **maxlength**
Allows to specify the maximum number of characters a user can enter into the text box.

Example

```
<form >
  First name: <input type = "text" name = "first_name" />
  <br>
  Last name: <input type = "text" name = "last_name" />
</form>
```

Password input controls

This is also a single-line text input but it masks the character as soon as a user enters it. They are also created using HTML <input>tag but type attribute is set to **password**.

Attributes

Sr.No	Attribute & Description
	type
1	Indicates the type of input control and for password input control it will be set to password .
	name
2	Used to give a name to the control which is sent to the server to be recognized and get the value.
	value
3	This can be used to provide an initial value inside the control.
4	size

Allows to specify the width of the text-input control in terms of characters.

maxlength

- 5 Allows to specify the maximum number of characters a user can enter into the text box.

Example

```
<form >
  User ID : <input type = "text" name = "user_id" />
  <br>
  Password: <input type = "password" name = "password" />
</form>
```

Multiple-Line Text Input Controls

This is used when the user is required to give details that may be longer than a single sentence. Multi-line input controls are created using HTML <textarea> tag.

Attributes

Sr.No	Attribute & Description
	name
1	Used to give a name to the control which is sent to the server to be recognized and get the value.
	rows
2	Indicates the number of rows of text area box.
	cols
3	Indicates the number of columns of text area box

Example

```
<form>
  Description : <br />
  <textarea rows = "5" cols = "50" name = "description">
    Enter description here...
  </textarea>
</form>
```

Checkbox Control

Checkboxes are used when more than one option is required to be selected. They are also created using HTML `<input>` tag but type attribute is set to **checkbox**.

Attributes

Sr.No	Attribute & Description
	type
1	Indicates the type of input control and for checkbox input control it will be set to checkbox .
	name
2	Used to give a name to the control which is sent to the server to be recognized and get the value.
	value
3	The value that will be used if the checkbox is selected.
	checked
4	Set to <i>checked</i> if you want to select it by default.

Example

```
<form>
  <input type = "checkbox" name = "maths" value = "on"> Maths
  <input type = "checkbox" name = "physics" value = "on"> Physics
</form>
```

Radio Button Control

Radio buttons are used when out of many options, just one option is required to be selected. They are also created using HTML `<input>` tag but type attribute is set to **radio**.

Attributes

Following is the list of attributes for radio button.

Sr.No	Attribute & Description
1	type

Indicates the type of input control and for checkbox input control it will be set to radio.

name

- 2 Used to give a name to the control which is sent to the server to be recognized and get the value.

value

- 3 The value that will be used if the radio box is selected.

checked

- 4 Set to *checked* if you want to select it by default.

Example

```
<form>
  <input type = "radio" name = "subject" value = "maths"> Maths
  <input type = "radio" name = "subject" value = "physics"> Physics
</form>
```

Select Box Control

A select box, also called drop down box which provides option to list down various options in the form of drop down list, from where a user can select one or more options.

Attributes

Following is the list of important attributes of <select> tag –

Sr.No	Attribute & Description
	name
1	Used to give a name to the control which is sent to the server to be recognized and get the value.
	size
2	This can be used to present a scrolling list box.
	multiple
3	If set to "multiple" then allows a user to select multiple items from the menu.

Following is the list of important attributes of <option> tag –

Sr.No	Attribute & Description
	value
1	The value that will be used if an option in the select box box is selected.
	selected
2	Specifies that this option should be the initially selected value when the page loads.
	label
3	An alternative way of labeling options

Example

```
<select name = "dropdown">
  <option value = "Maths" selected>Maths</option>
  <option value = "Physics">Physics</option>
</select>
```

File Upload Box

If you want to allow a user to upload a file to your web site, you will need to use a file upload box, also known as a file select box. This is also created using the <input> element but type attribute is set to **file**.

Attributes

Sr.No	Attribute & Description
	name
1	Used to give a name to the control which is sent to the server to be recognized and get the value.
	accept
2	Specifies the types of files that the server accepts.

Example

```
<form>
  <input type = "file" name = "fileupload" accept = "image/*" />
</form>
```

Button Controls

There are various ways in HTML to create clickable buttons. You can also create a clickable button using `<input>` tag by setting its type attribute to **button**. The type attribute can take the following values –

Sr.No	Type & Description
1	submit This creates a button that automatically submits a form.
2	reset This creates a button that automatically resets form controls to their initial values.
3	button This creates a button that is used to trigger a client-side script when the user clicks that button.
4	image This creates a clickable button but we can use an image as background of the button.

Example

```
<form>
  <input type = "submit" name = "submit" value = "Submit" />
  <input type = "reset" name = "reset" value = "Reset" />
  <input type = "button" name = "ok" value = "OK" />
  <input type = "image" name = "imagebutton" src =
"/html/images/logo.png" />
</form>
```

Hidden Form Controls

Hidden form controls are used to hide data inside the page which later on can be pushed to the server. This control hides inside the code and does not appear on the actual page. For example, following hidden form is being used to keep current page number. When a user will click next page then the value of hidden control will be sent to the web server and there it will decide which page will be displayed next based on the passed current page.

Example

```
<form>
  <p>This is page 10</p>
  <input type = "hidden" name = "pagename" value = "10" />
  <input type = "submit" name = "submit" value = "Submit" />
  <input type = "reset" name = "reset" value = "Reset" />
</form>
```

HTML5 – Overview

HTML5 tutorial provides details of all 40+ HTML tags including audio, video, header, footer, data, datalist, article etc. This HTML tutorial is designed for beginners and professionals.

HTML5 is a next version of HTML. Here, you will get some brand new features which will make HTML much easier. These new introducing features make your website layout clearer to both website designers and users. There are some elements like <header>, <footer>, <nav> and <article> that define the layout of a website.

Why use HTML5

It is enriched with advance features which makes it easy and interactive for designer/developer and users.

It allows you to play a video and audio file.

It allows you to draw on a canvas.

It facilitate you to design better forms and build web applications that work offline.

It provides you advance features for that you would normally have to write JavaScript to do.

HTML 5 Example

```
<!DOCTYPE>
<html>
<body>
<h1>Write Your First Heading</h1>
<p>Write Your First Paragraph.</p>
</body>
</html>
```

HTML5 - Syntax

The HTML 5 language has a "custom" HTML syntax that is compatible with HTML 4 and XHTML1 documents published on the Web, but is not compatible with the more esoteric SGML features of HTML 4.

HTML 5 does not have the same syntax rules as XHTML where we needed lower case tag names, quoting our attributes, an attribute had to have a value and to close all empty elements.

HTML5 comes with a lot of flexibility and it supports the following features –

- Uppercase tag names.
- Quotes are optional for attributes.
- Attribute values are optional.
- Closing empty elements are optional.

The DOCTYPE

DOCTYPEs in older versions of HTML were longer because the HTML language was SGML based and therefore required a reference to a DTD.

HTML 5 authors would use simple syntax to specify DOCTYPE as follows –

```
<!DOCTYPE html>
```

The above syntax is case-insensitive.

Character Encoding

HTML 5 authors can use simple syntax to specify Character Encoding as follows –

```
<meta charset = "UTF-8">
```

The above syntax is case-insensitive.

The <script> tag

It's common practice to add a type attribute with a value of "text/javascript" to script elements as follows

```
<script type = "text/javascript" src = "scriptfile.js"></script>
```

HTML 5 removes extra information required and you can use simply following syntax –

```
<script src = "scriptfile.js"></script>
```

The <link> tag

So far you were writing <link> as follows –

```
<link rel = "stylesheet" type = "text/css" href = "stylefile.css">
```

HTML 5 removes extra information required and you can simply use the following syntax –

```
<link rel = "stylesheet" href = "stylefile.css">
```

HTML5 Elements

HTML5 elements are marked up using start tags and end tags. Tags are delimited using angle brackets with the tag name in between.

The difference between start tags and end tags is that the latter includes a slash before the tag name.

Following is the example of an HTML5 element –

```
<p>...</p>
```

HTML5 tag names are case insensitive and may be written in all uppercase or mixed case, although the most common convention is to stick with lowercase.

Most of the elements contain some content like <p>...</p> contains a paragraph. Some elements, however, are forbidden from containing any content at all and these are known as void elements. For example, **br**, **hr**, **link**, **meta**, etc.

HTML5 Attributes

Elements may contain attributes that are used to set various properties of an element.

Some attributes are defined globally and can be used on any element, while others are defined for specific elements only. All attributes have a name and a value and look like as shown below in the example.

Following is the example of an HTML5 attribute which illustrates how to mark up a div element with an attribute named class using a value of "example" –

```
<div class = "example">...</div>
```

Attributes may only be specified within start tags and must never be used in end tags.

HTML5 attributes are case insensitive and may be written in all uppercase or mixed case, although the most common convention is to stick with lowercase.

HTML5 Document

The following tags have been introduced for better structure –

- **section** – This tag represents a generic document or application section. It can be used together with h1-h6 to indicate the document structure.
- **article** – This tag represents an independent piece of content of a document, such as a blog entry or newspaper article.
- **aside** – This tag represents a piece of content that is only slightly related to the rest of the page.
- **header** – This tag represents the header of a section.
- **footer** – This tag represents a footer for a section and can contain information about the author, copyright information, et cetera.
- **nav** – This tag represents a section of the document intended for navigation.
- **dialog** – This tag can be used to mark up a conversation.
- **figure** – This tag can be used to associate a caption together with some embedded content, such as a graphic or video.

The markup for an HTML 5 document would look like the following –

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<meta charset = "utf-8">
```

```
<title>...</title>
```

```
</head>
```

```
<body>
```

```
<header>...</header>
```

```
<nav>...</nav>
```

```
<article>
```

```
<section>
```

```

    ...
    </section>
</article>
<aside>...</aside>

<footer>...</footer>
</body>
</html>

```

Example:

```

<!DOCTYPE html>

<html>
  <head>
    <meta charset = "utf-8">
    <title>...</title>
  </head>

  <body>
    <header role = "banner">
      <h1>HTML5 Document Structure Example</h1>
      <p>This page should be tried in safari, chrome or Mozilla.</p>
    </header>

    <nav>
      <ul>
        <li><a href = "https://www.tutorialspoint.com/html">HTML
Tutorial</a></li>
        <li><a href = "https://www.tutorialspoint.com/css">CSS
Tutorial</a></li>
        <li><a href = "https://www.tutorialspoint.com/javascript">
JavaScript Tutorial</a></li>
      </ul>
    </nav>

    <article>
      <section>
        <p>Once article can have multiple sections</p>
      </section>
    </article>

```

```

<aside>
  <p>This is aside part of the web page</p>
</aside>

<footer>
  <p>Created by <a href = "https://tutorialspoint.com/">Tutorials
Point</a></p>
</footer>

</body>
</html>

```

HTML5 - Attributes

The placeholder attribute

HTML5 introduced a new attribute called **placeholder**. This attribute on `<input>` and `<textarea>` elements provide a hint to the user of what can be entered in the field. The placeholder text must not contain carriage returns or line-feeds.

Here is the simple syntax for placeholder attribute –

```
<input type = "text" name = "search" placeholder = "search the web"/>
```

The autofocus attribute

This is a simple one-step pattern, easily programmed in JavaScript at the time of document load, automatically focus one particular form field.

HTML5 introduced a new attribute called **autofocus** which would be used as follows –

```
<input type = "text" name = "search" autofocus/>
```

The required attribute

Now you do not need to have JavaScript for client-side validations like empty text box would never be submitted because HTML5 introduced a new attribute called **required** which would be used as follows and would insist to have a value –

```
<input type = "text" name = "search" required/>
```

HTML5 - Web Forms

The <input> element in HTML5

Apart from the above-mentioned attributes, HTML5 input elements introduced several new values for the **type** attribute. These are listed below.

Sr.No.	Type & Description
	<u>datetime</u>
1	A date and time (year, month, day, hour, minute, second, fractions of a second) encoded according to ISO 8601 with the time zone set to UTC.
	<u>datetime-local</u>
2	A date and time (year, month, day, hour, minute, second, fractions of a second) encoded according to ISO 8601, with no time zone information.
	<u>date</u>
3	A date (year, month, day) encoded according to ISO 8601.
	<u>month</u>
4	A date consisting of a year and a month encoded according to ISO 8601.
	<u>week</u>
5	A date consisting of a year and a week number encoded according to ISO 8601.
	<u>time</u>
6	A time (hour, minute, seconds, fractional seconds) encoded according to ISO 8601.
	<u>number</u>
7	It accepts only numerical value. The step attribute specifies the precision, defaulting to 1.
8	<u>range</u>

The range type is used for input fields that should contain a value from a range of numbers.

email

- 9 It accepts only email value. This type is used for input fields that should contain an e-mail address. If you try to submit a simple text, it forces to enter only email address in email@example.com format.

url

- 10 It accepts only URL value. This type is used for input fields that should contain a URL address. If you try to submit a simple text, it forces to enter only URL address either in http://www.example.com format or in http://example.com format.

HTML5 - Audio & Video

HTML5 features include native audio and video support without the need for Flash.

The HTML5 <audio> and <video> tags make it simple to add media to a website. You need to set **src** attribute to identify the media source and include a controls attribute so the user can play and pause the media.

Embedding Video

Here is the simplest form of embedding a video file in your webpage –

```
<video src = "foo.mp4" width = "300" height = "200" controls>
```

Your browser does not support the <video> element.

```
</video>
```

You can use <source> tag to specify media along with media type and many other attributes. A video element allows multiple source elements and browser will use the first recognized format –

```
<video width = "300" height = "200" controls autoplay>
```

```
<source src = "/html5/foo.ogg" type = "video/ogg" />
```

```
<source src = "/html5/foo.mp4" type = "video/mp4" />
```

Your browser does not support the <video> element.

```
</video>
```

Video Attribute

Sr.No.	Attribute & Description
	Autoplay
1	This Boolean attribute if specified, the video will automatically begin to play back as soon as it can do so without stopping to finish loading the data.
	autobuffer
2	This Boolean attribute if specified, the video will automatically begin buffering even if it's not set to automatically play.
	controls
3	If this attribute is present, it will allow the user to control video playback, including volume, seeking, and pause/resume playback.
	height
4	This attribute specifies the height of the video's display area, in CSS pixels.
	loop
5	This Boolean attribute if specified, will allow video automatically seek back to the start after reaching at the end.
	preload
6	This attribute specifies that the video will be loaded at page load, and ready to run. Ignored if autoplay is present.
	poster
7	This is a URL of an image to show until the user plays or seeks.
	src
8	The URL of the video to embed. This is optional; you may instead use the <source> element within the video block to specify the video to embed.
9	width

This attribute specifies the width of the video's display area, in CSS pixels.

Embedding Audio

HTML5 supports <audio> tag which is used to embed sound content in an HTML or XHTML document as follows.

```
<audio src = "foo.wav" controls autoplay>
```

Your browser does not support the <audio> element.

```
</audio>
```

You can use <source> tag to specify media along with media type and many other attributes. An audio element allows multiple source elements and browser will use the first recognized format -

```
<audio controls autoplay>
```

```
<source src = "/html5/audio.ogg" type = "audio/ogg" />
```

```
<source src = "/html5/audio.wav" type = "audio/wav" />
```

Your browser does not support the <audio> element.

```
</audio>
```

Audio Attribute

Sr.No.	Attribute & Description
	autoplay
1	This Boolean attribute if specified, the audio will automatically begin to play back as soon as it can do so without stopping to finish loading the data.
	autobuffer
2	This Boolean attribute if specified, the audio will automatically begin buffering even if it's not set to automatically play.
	controls
3	If this attribute is present, it will allow the user to control audio playback, including volume, seeking, and pause/resume playback.
	loop
4	This Boolean attribute if specified, will allow audio automatically seek back to the start after reaching at the end.

preload

- 5 This attribute specifies that the audio will be loaded at page load, and ready to run. Ignored if autoplay is present.

src

- 6 The URL of the audio to embed. This is optional; you may instead use the <source> element within the video block to specify the video to embed.

Handling Media Events

The HTML5 audio and video tag can have a number of attributes to control various functionalities of the control using JavaScript –

S.No.	Event & Description
	Abort
1	This event is generated when playback is aborted.
	Canplay
2	This event is generated when enough data is available that the media can be played.
	Ended
3	This event is generated when playback completes.
	Error
4	This event is generated when an error occurs.
	Loadeddata
5	This event is generated when the first frame of the media has finished loading.
	loadstart
6	This event is generated when loading of the media begins.
	pause
7	This event is generated when playback is paused.

8 play

This event is generated when playback starts or resumes.

progress

9 This event is generated periodically to inform the progress of the downloading the media.

10 ratechange

This event is generated when the playback speed changes.

11 sought

This event is generated when a seek operation completes.

12 seeking

This event is generated when a seek operation begins.

13 Suspend

This event is generated when loading of the media is suspended.

14 Volumechange

This event is generated when the audio volume changes.

Waiting

15 This event is generated when the requested operation (such as playback) is delayed pending the completion of another operation (such as a seek).

```
<!DOCTYPE HTML>
```

```
<html>
```

```
<head>
```

```
<script type = "text/javascript">
```

```
function PlayVideo() {
```

```
    var v = document.getElementsByTagName("video")[0];
```

```
    v.play();
```

```
}
```

```
</script>
```

```
</head>

<body>

  <form>
    <video width = "300" height = "200" src = "/html5/foo.mp4">
      Your browser does not support the video element.
    </video>
    <br />
    <input type = "button" onclick = "PlayVideo();" value = "Play"/>
  </form>

</body>
</html>
```